

The Wasserstein Distance

When $c(x, y) = d(x, y)^p, p \geq 1$, Kantorovich OT is called **Wasserstein Distance**

$$W_p(\alpha, \beta) = \left(\inf_{\pi \in \Pi(\alpha, \beta)} \int_X c(x, y)^p \, d\pi(x, y) \right)^{1/p}$$

Next Time

- The Wasserstein distance(s)
- Metric properties of Optimal Transport
- Dual problem
- Reading: Peyré & Cuturi (2019), Sections 2.4 - 2.6